

# Muhamad Assem Ahmed

## Lab 19

### Part A

LAB 19 — Lab ID: 27 | MIN\_GRADE\_LAB = 2.5 | COURSE\_COUNT = 2

Index only displays Enrollments.Count which just needs the Enrollment rows themselves loaded  
Details additionally prints each enrolled student's name, which is further

### Part B

StudentPortalWeb Home Privacy

## Enroll a Student

Links one real Student to one real Course. Everything typed here is sent to the POST version of **create**.

Student

Kareem Fouad  
Nada Samir  
Omar Hesham  
Yara Adel

Grade (leave blank if not yet graded)

EnrollCancel

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## Course

Database Fundamentals  
Web Development Using .NET

EnrollCancel

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### Part C

A hidden field is still client-supplied data anyone can edit the HTML or replay the request with a different value

StudentPortalWeb Home Privacy

## Omar Hesham

Reached through the route pattern **students/{id:int}**.

|                      |             |
|----------------------|-------------|
| <b>Id</b>            | 3           |
| <b>Full Name</b>     | Omar Hesham |
| <b>Year of Study</b> | 3           |
| <b>GPA</b>           | 2.80        |

### Enrolled Courses

| Course                     | Enrolled On | Grade          |
|----------------------------|-------------|----------------|
| Web Development Using .NET | 2026-08-05  | Not yet graded |

[Back to all students](#)

Detail Pages are reachable directly by id - try `/students/1` through `/students/5`

Part D

StudentPortalWeb Home Privacy

Enroll a Student

Links one real Student to one real Course. Everything typed here is sent to the POST version of `create`.

Student

Yara Adel

Course

Database Fundamentals

Grade (leave blank if not yet graded)

2

Grade must be at least 2.5

Enroll

Cancel

StudentPortalWeb Home Privacy

Yara Adel

Reached through the route pattern `students/{id:int}`.

|               |           |
|---------------|-----------|
| Id            | 4         |
| Full Name     | Yara Adel |
| Year of Study | 2         |
| GPA           | 3.50      |

Enrolled Courses

| Course                     | Enrolled On | Grade        |
|----------------------------|-------------|--------------|
| Web Development Using .NET | 2026-08-05  | 3.95 - First |
| Database Fundamentals      | 2026-08-05  | 2.50 - Pass  |

[Back to all students](#)

Detail Pages are reachable directly by id - try `/students/1` through `/students/5`

|   | Id | StudentId | CourseId | Enrollment...  | Grade |
|---|----|-----------|----------|----------------|-------|
| ▶ | 5  | 4         | 2        | 2026-08-05 ... | 3.95  |
|   | 7  | 3         | 2        | 2026-08-05 ... | NULL  |
|   | 8  | 2         | 3        | 2026-08-05 ... | 2     |
|   | 9  | 4         | 3        | 2026-08-05 ... | 2.5   |

Part E

StudentPortalWeb Home Privacy

Muhamad Assem

Reached through the route pattern `students/{id:int}`.

|               |               |
|---------------|---------------|
| Id            | 5             |
| Full Name     | Muhamad Assem |
| Year of Study | 4             |
| GPA           | 3.70          |

Enrolled Courses

| Course                | Enrolled On | Grade          |
|-----------------------|-------------|----------------|
| Database Fundamentals | 2026-08-05  | Not yet graded |

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Detail Pages are reachable directly by id - try `/students/1` through `/students/5`

StudentPortalWeb Home Privacy

Muhamad Assem

Reached through the route pattern `students/{id:int}`.

|               |               |
|---------------|---------------|
| Id            | 5             |
| Full Name     | Muhamad Assem |
| Year of Study | 4             |
| GPA           | 3.70          |

Enrolled Courses

| Course                     | Enrolled On | Grade          |
|----------------------------|-------------|----------------|
| Database Fundamentals      | 2026-08-05  | Not yet graded |
| Web Development Using .NET | 2026-08-05  | 2.50 - Pass    |

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Detail Pages are reachable directly by id - try `/students/1` through `/students/5`

Exception User-Unhandled

Microsoft.EntityFrameworkCore.DbUpdateException: 'An error occurred while saving the entity changes. See the inner exception for details.'

Inner Exception

SqlException: Cannot insert duplicate key row in object 'dbo.Enrollments' with unique

This exception was originally thrown at this call stack:  
[External Code]

Analyze with Copilot | Show Call Stack | View Details | Copy Details

Exception Settings

☐ Break when this exception type is thrown

☒ Break when this exception type is user-unhandled

Except when thrown from:

☐ StudentPortalWeb.dll

Open Exception Settings | Edit Conditions

The second insert threw a DbUpdateException from SQLException

## Part E

Lab ID = 27

$\text{MIN\_GRADE\_LAB} = 1.0 + (27 \bmod 4) \times 0.5 = 1.0 + (3 \times 0.5) = 2.5$

$\text{COURSE\_COUNT} = (27 \bmod 3) + 2 = 0 + 2 = 2$

Enrollment data can be rejected in three places:

client-side, if the browser's own HTML5 validation catches an obviously bad value before the request is even sent

ModelState/[Range], server-side but still before the database, in the POST action's

the database itself, via constraints like the unique index

A single foreign key column can only ever point at one row, but Student and Course are each "many" to the other a student takes several courses and a course has several students. What actually fixes it is a third, associative entity

I'd choose Cascade. An Assignment has no independent meaning once its Course is deleted there's no such thing as an assignment for a course that no longer exists